Question ID: f2495de4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Easy

Question

What is the value of $\cos \frac{565\pi}{6}$?

Answer

A. $\frac{1}{2}$

B. 1

C. $\frac{\sqrt{3}}{2}$

D. $\sqrt{3}$

Correct Answer: C

Rationale

Choice C is correct. The cosine of an angle is equal to the cosine of $n(2\pi)$ radians more than the angle, where n is an integer constant. Since $\frac{565\pi}{6}$ is equivalent to $47(2\pi) + \frac{\pi}{6}$, $\cos\left(\frac{565\pi}{6}\right)$ can be rewritten as $\cos\left(47(2\pi) + \frac{\pi}{6}\right)$, which is equal to $\cos\left(\frac{\pi}{6}\right)$. Therefore, the value of $\cos\left(\frac{565\pi}{6}\right)$ is equal to the value of $\cos\left(\frac{\pi}{6}\right)$, which is $\frac{\sqrt{3}}{2}$.

Alternate approach: A trigonometric ratio can be found using the unit circle, that is, a circle with radius 1 unit. The cosine of a number t is the *x*-coordinate of the point resulting from traveling a distance of t counterclockwise from the point $(1, 0)$ around a unit circle centered at the origin in the *xy*-plane. A unit circle has a circumference of 2π. It follows that since $\frac{565\pi}{6}$ is equal to $47(2\pi) + \frac{\pi}{6}$, traveling a distance of $\frac{565\pi}{6}$ counterclockwise around a unit circle means traveling around the circle completely 47 times and then another $\frac{\pi}{6}$ beyond that. That is, traveling $\frac{565\pi}{6}$ results in the same point as traveling $\frac{\pi}{6}$. Traveling $\frac{\pi}{6}$ counterclockwise from the point $(1, 0)$ around a unit circle centered at the origin in the *xy*-plane results in the point $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. Thus, the value of $\cos \frac{565\pi}{6}$ is the *x*-coordinate of the point $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$, which is $\frac{\sqrt{3}}{2}$.

Choice A is incorrect. This is the value of $\sin \frac{565\pi}{6}$, not $\cos \frac{565\pi}{6}$.

Choice B is incorrect. This is the value of the cosine of a multiple of 2π, not $\frac{565\pi}{6}$.

Choice D is incorrect. This is the value of $\frac{1}{\tan \frac{565\pi}{6}}$, not $\cos \frac{565\pi}{6}$.